fun-miRBench Benchmark Report

Run setting	Value
Datasets evaluated	21
Datasets skipped	0
Predictors evaluated	5
Responsive genes	oriented shrunken log2FC > 0; FSOS s-value < 0.05 against d = 0.5
No-response genes	whole shrunken 95% interval of the oriented change below d = 0.5

Evaluation design

- Higher aligned scores mean stronger predicted targeting.

- Every predictor is scored on the same 19,226 genes; unscored genes rank last.

- Unresolved genes are left out of the AUROC.

- Overexpression (OE) datasets only.

Key results

- Highest mean AUROC: miRBind2-3UTR (0.643).

- Highest mean Spearman: miRBind2-3UTR (0.095).

- Highest coverage: miRBind2-3UTR (98.3%).

- Lowest coverage: miRDB (14.0%).

Cross-Dataset Predictor Summary

Coverage notes

- Coverage is score availability; a gene a predictor does not score ranks last.

- AUROC on 17 of 21 datasets (at least 30 responsive and 30 no-response genes).

- Spearman on all 21 datasets.

- Low coverage (<25%): TargetScan v8: 16.6%; miRDB: 14.0%.

Predictor	Coverage	AUROC	Spearman	Decoy z-score
TargetScan v8	16.6%	0.627	0.078	3.169
miRDB	14.0%	0.632	0.087	3.478
microT-CNN	90.7%	0.610	0.040	1.116
miRBind2-3UTR	98.3%	0.643	0.095	2.827
miRAW	92.9%	0.574	0.012	0.194

Dataset Inventory

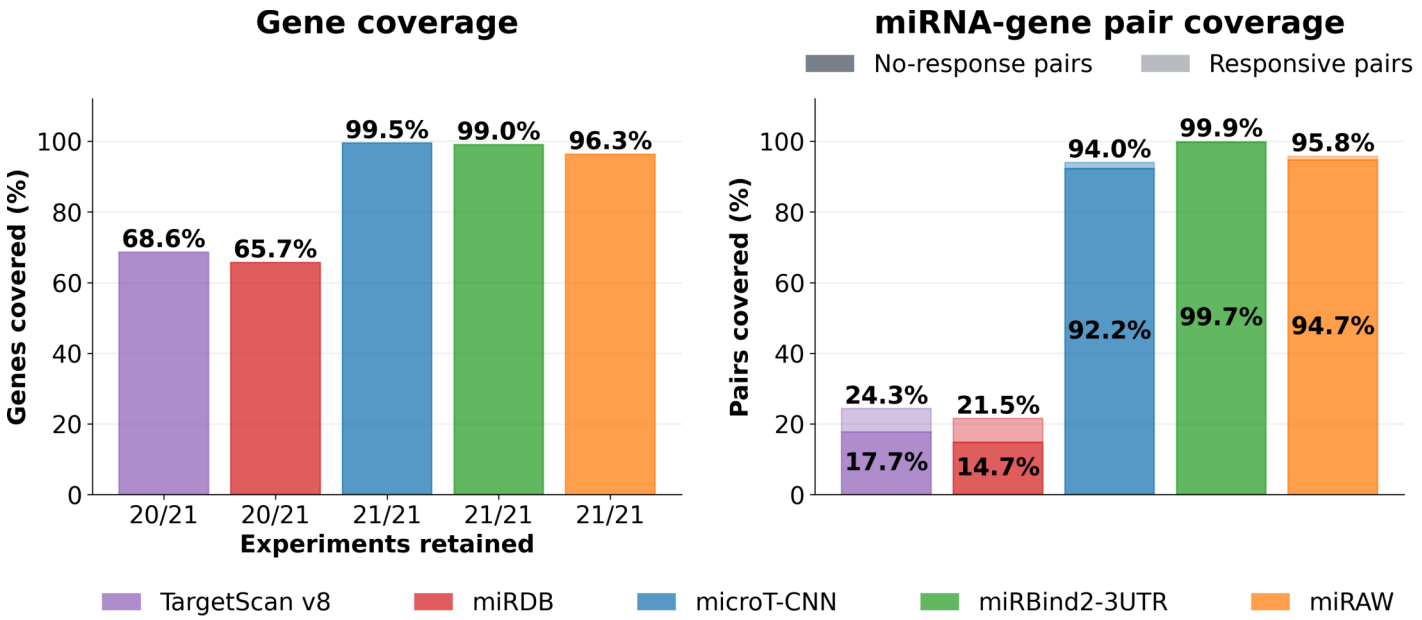
Experiment	Experiment ID
hsa-miR-574-5p · MGC-803	GSE112859_OE_miR_574_5p
hsa-miR-101-3p · HepG2	GSE115646_OE_miR-101-3p_HepG2
hsa-miR-133a-3p · HepG2	GSE115646_OE_miR-133a-3p_HepG2
hsa-miR-140-3p · HepG2	GSE115646_OE_miR-140-3p_HepG2
hsa-miR-140-5p · HepG2	GSE115646_OE_miR-140-5p_HepG2
hsa-miR-18a-5p · HepG2	GSE115646_OE_miR-18a-5p_HepG2
hsa-miR-211-5p · HepG2	GSE115646_OE_miR-211-5p_HepG2
hsa-miR-150-5p · LX-2	GSE124411_OE_miR_150_5p_1
hsa-miR-450a-5p · A2780	GSE129076_OE_miR_450a_5p_1
hsa-miR-450a-5p · SKOV-3	GSE129076_OE_miR_450a_5p_2
hsa-miR-1271-5p · SH-SY5Y	GSE148827_OE_miR_1271_5p
hsa-miR-21-3p · HaCaT	GSE154968_OE_miR_21_3p
hsa-miR-20b-5p · HepG2	GSE168484_OE_miR_20b_5p
hsa-miR-100-3p · Kelly	GSE175729_OE_miR-100-3p_Kelly
hsa-miR-1915-3p · Kelly	GSE175729_OE_miR-1915-3p_Kelly
hsa-miR-380-3p · Kelly	GSE175729_OE_miR-380-3p_Kelly
hsa-miR-485-3p · Kelly	GSE175729_OE_miR-485-3p_Kelly
hsa-miR-99b-5p · Kelly	GSE175729_OE_miR-99b-5p_Kelly
hsa-miR-130a-3p · bone-marrow mesenchymal stem cells	GSE202135_OE_miR_130a_3p
hsa-miR-130b-3p · bone-marrow mesenchymal stem cells	GSE202135_OE_miR_130b_3p
hsa-miR-375-3p · HUV-EC-C	GSE210778_OE_miR_375_3p

Coverage

- Gene coverage: share of measured genes the tool scores in at least one experiment.

- Experiments retained: experiments where it scores responsive and no-response genes.

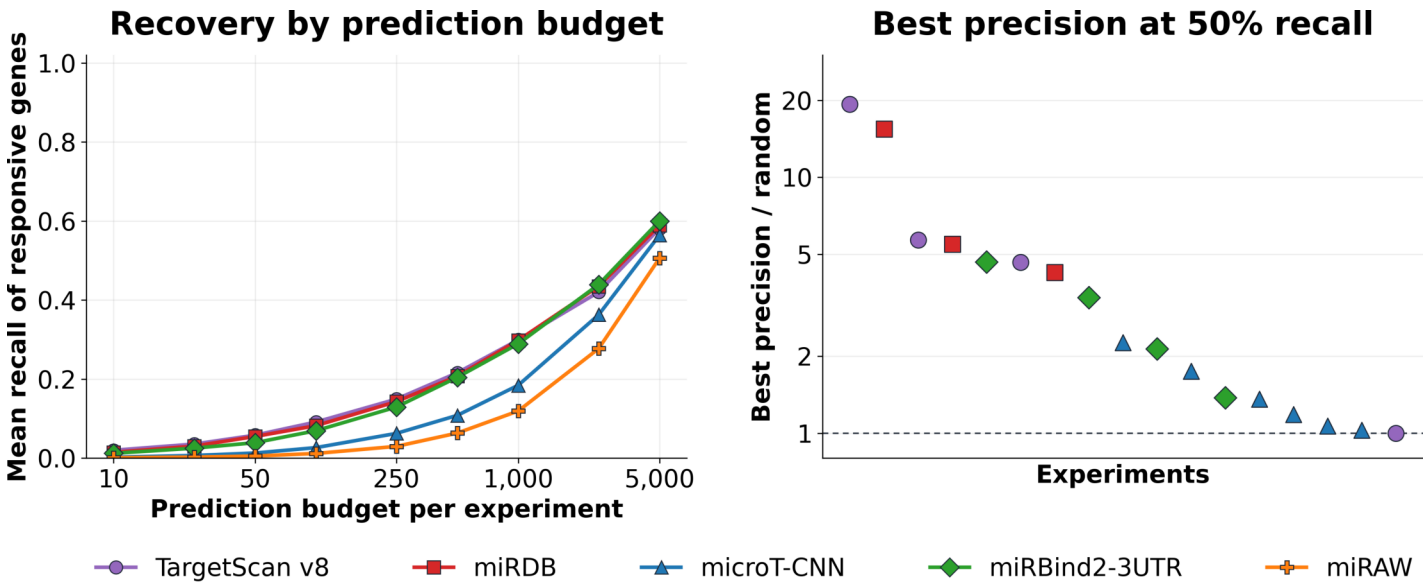
- Pair coverage: share of responsive and of no-response miRNA-gene pairs it scores.



Recovery

- Recall: share of responsive genes among the top-ranked genes, mean over experiments.

- Best precision / random: best tool's precision at 50% recall over responsive share.



Performance

- AUROC: chance that a responsive gene scores above a no-response gene; 0.5 is chance.

- Spearman rho: agreement of the score's ranking with the shrunken log2FC; 0 is none.

- Decoy z-score: Spearman for the tested miRNA against different-seed miRNAs, in SDs.

- Experiments won: experiments where the tool has the highest AUROC or Spearman rho.

